Rainfall Prediction from Cloud Images using Deep learning

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General Objective

- Develop intelligent rainfall prediction systems using Deep Learning and Machine Learning
- Apply Transfer Learning, Feature Extraction, Sequential Learning, and Multimodal Fusion
- Compare spatial, temporal, and multimodal rainfall prediction approaches
- Evaluate models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC
- Compare strengths and limitations of different rainfall prediction approaches

Approach 1: Ground-Based Multimodal Rainfall Prediction

Subhadeep Barman (2022ITB048)

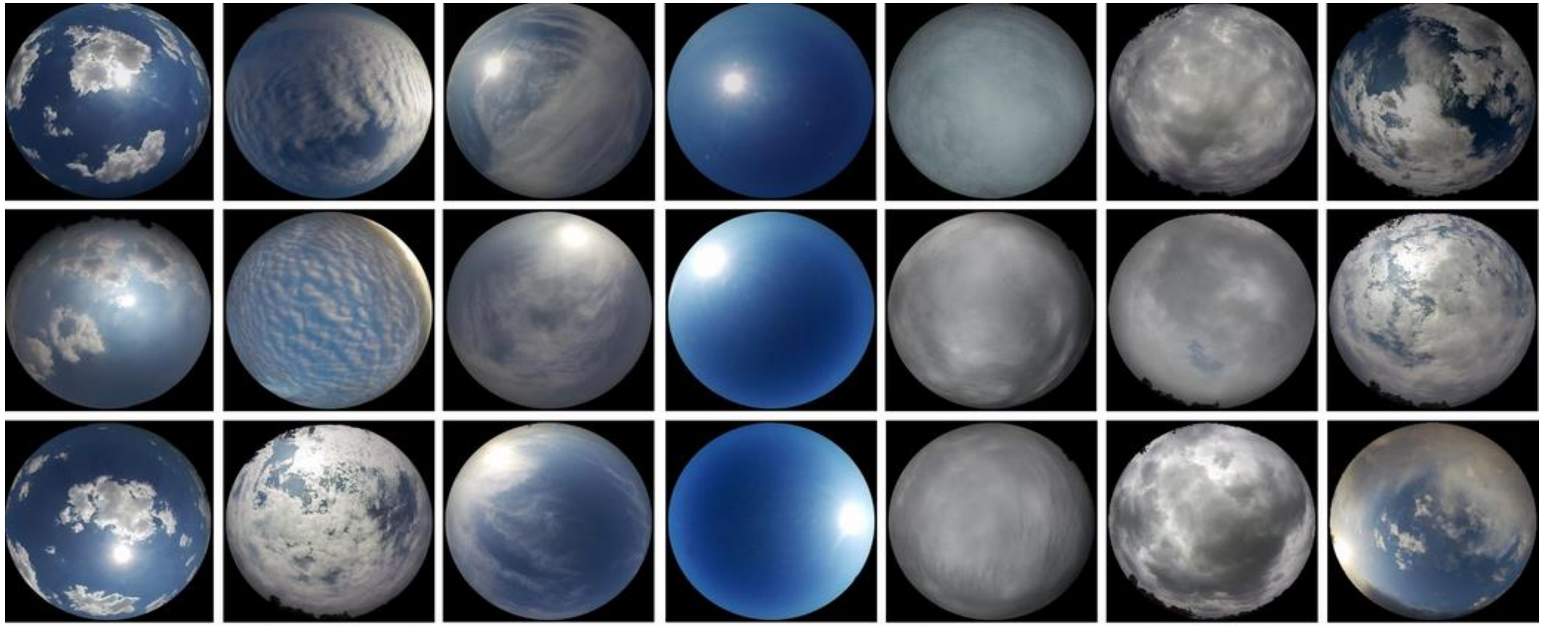
Dataset - MGCD

- **Multimodal Ground-Based Cloud Database** (MGCD): Collected in Tianjin, China (2017 – 2018) - Total Samples: 8000 (Format: JPEG)
- Captured using **fisheye sky camera** (Resolution: 1024 × 1024)
- Each image paired with corresponding **meteorological observations**
- 7 cloud types: Converted into **Binary Labels** (Rain vs No-Rain) for our task
- Rain vs No-Rain labels were derived based on: **Meteorological domain knowledge**

- **Labeling Logic:**

  Rain Class > Clouds typically associated with precipitation (Cb and Ns)

  No-Rain Class > Non-precipitating and clear sky conditions (Others)



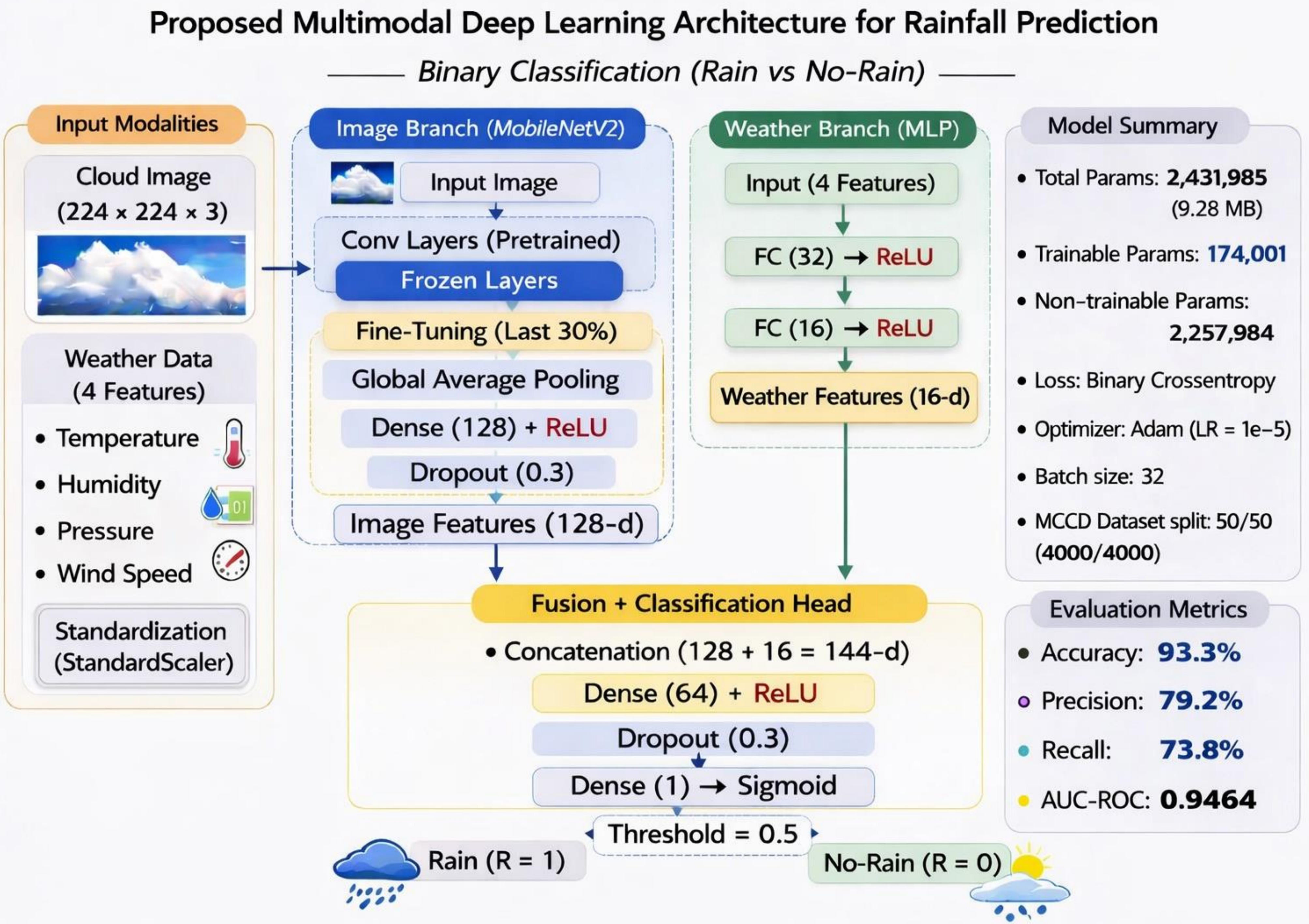
Hereunder, the table lists the corresponding multimodal cloud information of the above ground-based cloud images.

Type	Cumulus	Altocumulus and cirrocumulus	Cirrus and cirrostratus	Clear sky	Stratocumulus, stratus and altostratus	Cumulonimbus and nimbostratus	Mixed cloud
Temp. (℃) Hum. (%RH) Pre. (hpa) W. Speed (m/s)	31 22.9 1018 0	31.6 59.2 1004.8 0	23.6 21.3 1014.5 0.6	31.2 35.4 1007.6 1.2	27.6 66.7 1006.6 0.3	35.4 61.4 1004.7 0	37.5 56.7 1004.9 0
Temp. (℃) Hum. (%RH) Pre. (hpa) W. Speed (m/s)	32.6 67.2 1005.9 1.7	33.8 30.7 1007.8 1.7	20.6 31.6 1017 0	29.2 45.3 1006.4 1.7	23 21.9 1014.7 0	14.1 30.6 1027.3 1.9	29.9 61.3 1009.5 2.4
Temp. (℃) Hum. (%RH) Pre. (hpa) W. Speed (m/s)	34 61.9 1005.5 0.5	24.2 22.9 1012 1.3	20.5 14.8 1010.7 2	22.6 20.1 1017 2.4	34.7 59.2 1004 0	25.8 99.5 1005.7 0.8	36.4 59.8 1004.9 1.7

```
/content/MGCD:
MGCD

/content/MGCD/MGCD:
test
train

/content/MGCD/MGCD/test:
1_cumulus
1_cumulus.xlsx
2_altocumulus
2_altocumulus.xlsx
3_cirrus
3_cirrus.xlsx
4_clearsky
4_clearsky.xlsx
5_stratocumulus
5_stratocumulus.xlsx
6_cumulonimbus
6_cumulonimbus.xlsx
7_mixed
7_mixed.xlsx

/content/MGCD/MGCD/test/1_cumulus:
1_cumulus_000691.jpg
1_cumulus_000692.jpg
1_cumulus_000693.jpg
1_cumulus_000694.jpg
1_cumulus_000695.jpg
1_cumulus_000696.jpg
```

	image_path	temp	humidity	pressure	wind	label
0	/content/MGCD/MGCD/test/2_altocumulus/2_altocu...	7.1	41.7	1031.8	0.7	0
1	/content/MGCD/MGCD/test/2_altocumulus/2_altocu...	6.4	43.2	1031.7	1.0	0
2	/content/MGCD/MGCD/test/2_altocumulus/2_altocu...	6.1	44.1	1031.6	0.5	0
3	/content/MGCD/MGCD/test/2_altocumulus/2_altocu...	5.6	45.1	1031.9	1.0	0
4	/content/MGCD/MGCD/test/2_altocumulus/2_altocu...	5.5	45.7	1031.9	0.3	0

	Number	Name	Temperature(℃)	Humidity(%RH)	Pressure(hpa)	Wind speed(m/s)
0	1	6_cumulonimbus_000001	27.3	56.4	1005.4	0.0
1	2	6_cumulonimbus_000002	27.3	57.0	1005.3	0.0
2	3	6_cumulonimbus_000003	26.8	87.1	1006.0	1.2
3	4	6_cumulonimbus_000004	26.8	87.5	1006.0	0.0
4	5	6_cumulonimbus_000005	26.7	87.3	1006.0	1.0



Proposed Multimodal Deep Learning Architecture for Rainfall Prediction
Binary Classification (Rain vs No-Rain)
Input Modalities
Cloud Image
(224 × 224 × 3)
Weather Data
(4 Features)
Temperature
Humidity
Pressure
Wind Speed
Standardization
(StandardScaler)
Image Branch (MobileNetV2)
Input Image
Conv Layers (Pretrained)
Frozen Layers
Fine-Tuning (Last 30%)
Global Average Pooling
Dense (128) + ReLU
Dropout (0.3)
Image Features (128-d)
Weather Branch (MLP)
Input (4 Features)
FC (32) → ReLU
FC (16) → ReLU
Weather Features (16-d)
Model Summary
Total Params: 2,431,985 (9.28 MB)
Trainable Params: 174,001
Non-trainable Params: 2,257,984
Loss: Binary Crossentropy
Optimizer: Adam (LR = 1e−5)
Batch size: 32
MCCD Dataset split: 50/50 (4000/4000)
Fusion + Classification Head
Concatenation (128 + 16 = 144-d)
Dense (64) + ReLU
Dropout (0.3)
Dense (1) → Sigmoid
Threshold = 0.5
Rain (R = 1)
No-Rain (R = 0)
Evaluation Metrics
Accuracy: 93.3%
Precision: 79.2%
Recall: 73.8%
AUC-ROC: 0.9464

Training Strategy

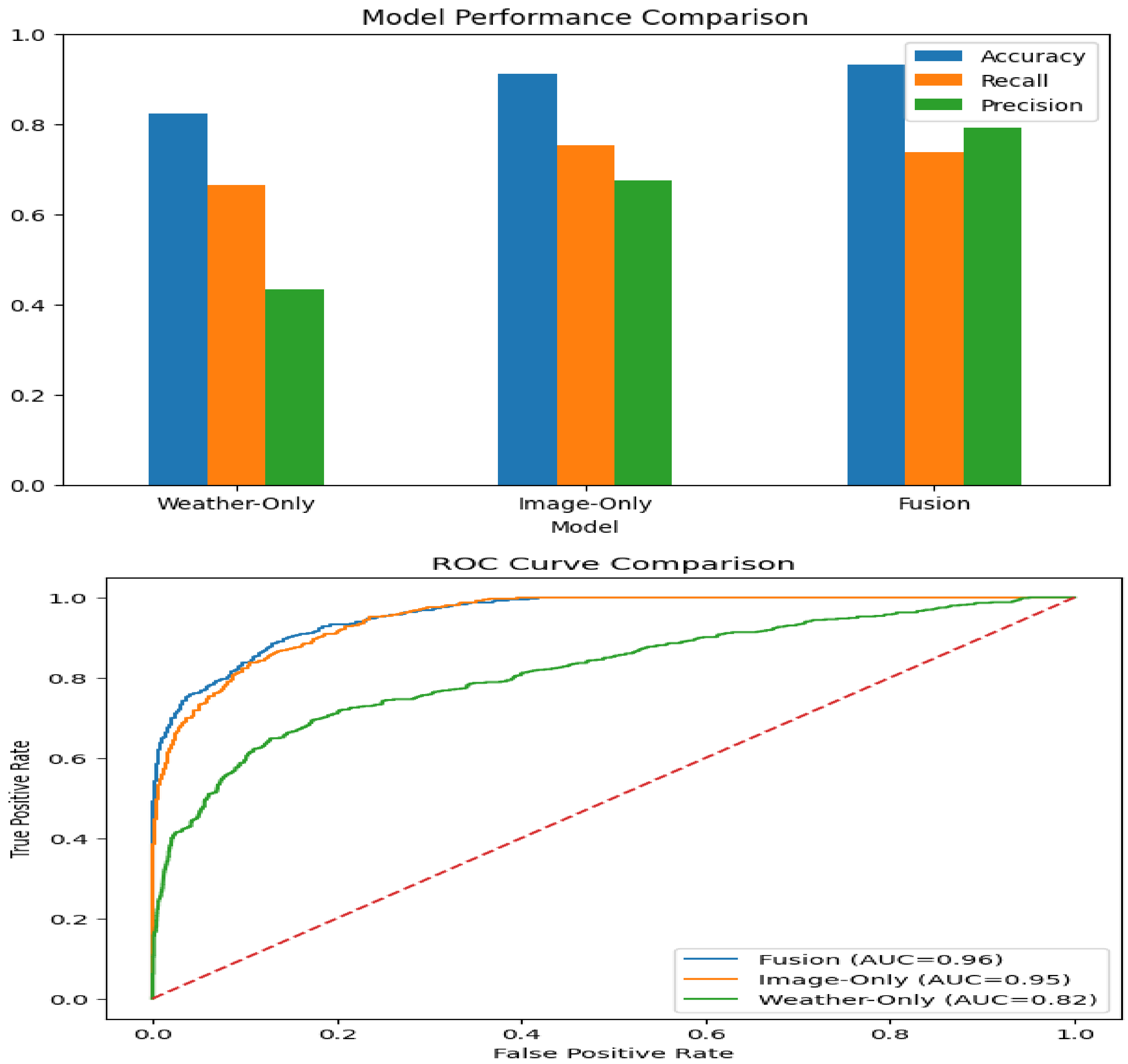
- **MobileNetV2** gives strong visual features with low computational cost and better generalization for medium-sized datasets
- **Transfer Learning** using **ImageNet pretrained weights**
- Applied **shuffling** during training to improve generalization
- **Early Stopping** to prevent overfitting
- **Learning rate scheduling** for stable fine-tuning

```
Epoch 1/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 408s 3s/step - accuracy: 0.7781 - auc: 0.6360 - loss: 0.4747 - precision: 0.2468 - recall: 0.1616 - val_accuracy: 0.8938 - val_auc: 0.9327 -
Epoch 2/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 381s 3s/step - accuracy: 0.9113 - auc: 0.9447 - loss: 0.2231 - precision: 0.8542 - recall: 0.5414 - val_accuracy: 0.9187 - val_auc: 0.9482 -
Epoch 3/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 404s 3s/step - accuracy: 0.9534 - auc: 0.9798 - loss: 0.1346 - precision: 0.8884 - recall: 0.7870 - val_accuracy: 0.9268 - val_auc: 0.9557 -
Epoch 4/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 392s 3s/step - accuracy: 0.9673 - auc: 0.9845 - loss: 0.1089 - precision: 0.9139 - recall: 0.8497 - val_accuracy: 0.9352 - val_auc: 0.9589 -
Epoch 5/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 389s 3s/step - accuracy: 0.9739 - auc: 0.9937 - loss: 0.0815 - precision: 0.9431 - recall: 0.8877 - val_accuracy: 0.9377 - val_auc: 0.9592 -
Epoch 6/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 363s 3s/step - accuracy: 0.9813 - auc: 0.9958 - loss: 0.0635 - precision: 0.9559 - recall: 0.9108 - val_accuracy: 0.9423 - val_auc: 0.9580 -
Epoch 7/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 408s 3s/step - accuracy: 0.9838 - auc: 0.9964 - loss: 0.0547 - precision: 0.9470 - recall: 0.9432 - val_accuracy: 0.9398 - val_auc: 0.9581 -
Epoch 8/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 364s 3s/step - accuracy: 0.9863 - auc: 0.9981 - loss: 0.0473 - precision: 0.9581 - recall: 0.9482 - val_accuracy: 0.9380 - val_auc: 0.9426 -
Epoch 9/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 371s 3s/step - accuracy: 0.9855 - auc: 0.9973 - loss: 0.0456 - precision: 0.9695 - recall: 0.9387 - val_accuracy: 0.9358 - val_auc: 0.9380 -
Epoch 10/10
125/125 ━━━━━━━━━━━━━━━━━━━━ 381s 3s/step - accuracy: 0.9908 - auc: 0.9991 - loss: 0.0327 - precision: 0.9852 - recall: 0.9552 - val_accuracy: 0.9395 - val_auc: 0.9428 -
```

```
Epoch 1/5
125/125 ━━━━━━━━━━━━━━━━━━━━ 492s 4s/step - accuracy: 0.9377 - auc: 0.9450 - loss: 0.1758 - precision: 0.8655 - recall: 0.6467 - val_accuracy: 0.9330 - val_auc: 0.9465 -
Epoch 2/5
125/125 ━━━━━━━━━━━━━━━━━━━━ 484s 3s/step - accuracy: 0.9871 - auc: 0.9953 - loss: 0.0535 - precision: 0.9703 - recall: 0.9440 - val_accuracy: 0.9317 - val_auc: 0.9434 -
Epoch 3/5
125/125 ━━━━━━━━━━━━━━━━━━━━ 442s 3s/step - accuracy: 0.9906 - auc: 0.9990 - loss: 0.0318 - precision: 0.9790 - recall: 0.9560 - val_accuracy: 0.9245 - val_auc: 0.9450 -
Epoch 4/5
125/125 ━━━━━━━━━━━━━━━━━━━━ 495s 3s/step - accuracy: 0.9943 - auc: 0.9995 - loss: 0.0237 - precision: 0.9837 - recall: 0.9797 - val_accuracy: 0.9325 - val_auc: 0.9397 -
Epoch 5/5
125/125 ━━━━━━━━━━━━━━━━━━━━ 445s 3s/step - accuracy: 0.9951 - auc: 0.9998 - loss: 0.0182 - precision: 0.9886 - recall: 0.9790 - val_accuracy: 0.9287 - val_auc: 0.9442 -
```

Model Comparison

Model	Accuracy	AUC	Recall (Rain)	Precision (Rain)
Fusion	0.9330	0.9464	0.7376	0.7916
Image-Only	0.9105	0.9508	0.7530	0.6748
Weather-Only	0.8225	0.8167	0.6644	0.4319



Model Performance Comparison
Accuracy
Recall
Precision
Weather-Only
Image-Only
Fusion
Model
1.0
0.8
0.6
0.4
0.2
0.0
ROC Curve Comparison
Fusion (AUC=0.96)
Image-Only (AUC=0.95)
Weather-Only (AUC=0.82)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

Limitations & Future Work

Limitations

- Binary classification only
- No temporal modelling
- Dataset from single region
- No rainfall intensity regression

Future Work

- Multi-class rainfall prediction
- Temporal modelling (CNN + LSTM)
- Multi-region cross validation
- Attention-based fusion

Approach 2: Satellite Image-Based Rainfall Classification

Ankesh Hatui (2022ITB040)

Dataset - LSCID-M

- Total Images: **104,790** (1000 x 1000)
- Used: **5,000** randomly sampled images
- **Goal:** Identify rainfall-related patterns (WJ class)

- **Classes:**

1. Earth Cloud (EC)
2. Fog/Smog (FS)
3. Thick Cloud (TC)
4. Water Jet (Rainfall)
5. Low Water Cloud (LWC)
6. High Ice Cloud (HIC)
7. Snow, Ocean, Desert, Vegetation

Feature Extraction

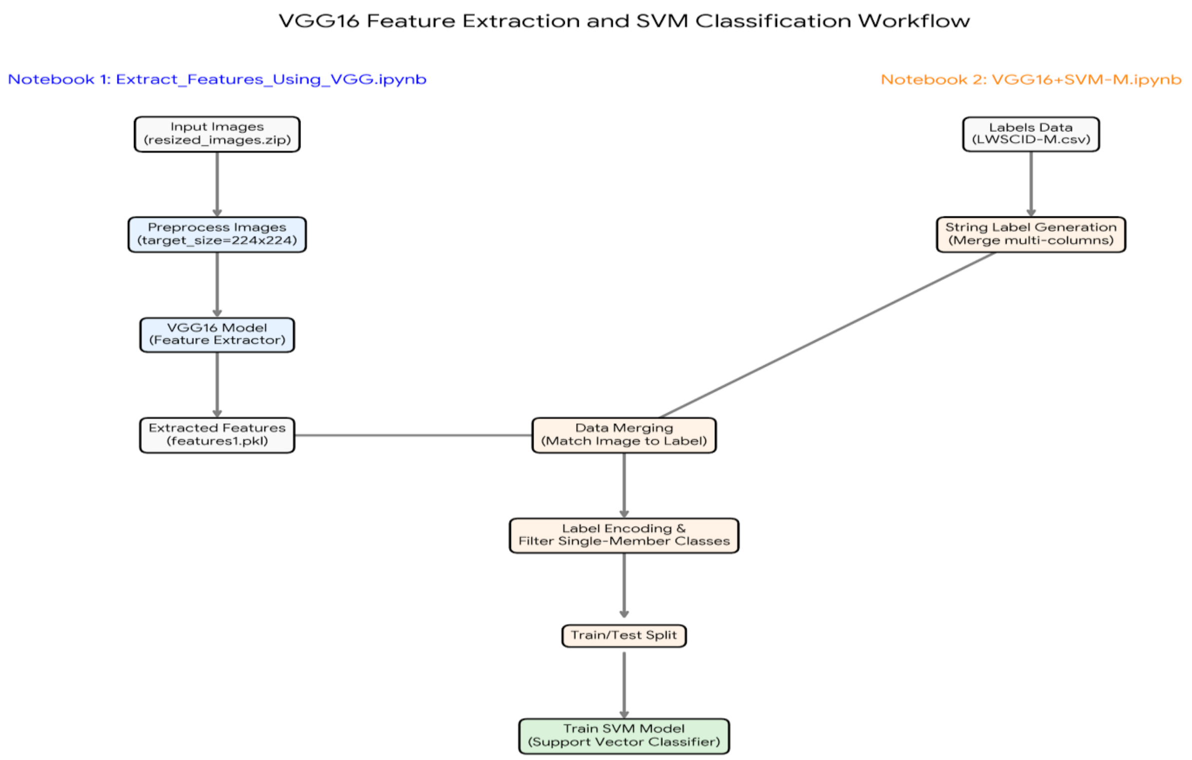
- **Method:** Extracted features using **VGG16** (pretrained CNN) and Stored in a **PKL file**
- **Process:** Input image → CNN → feature vector
- **Models Used**

1. EfficientNetB0
2. VGG16

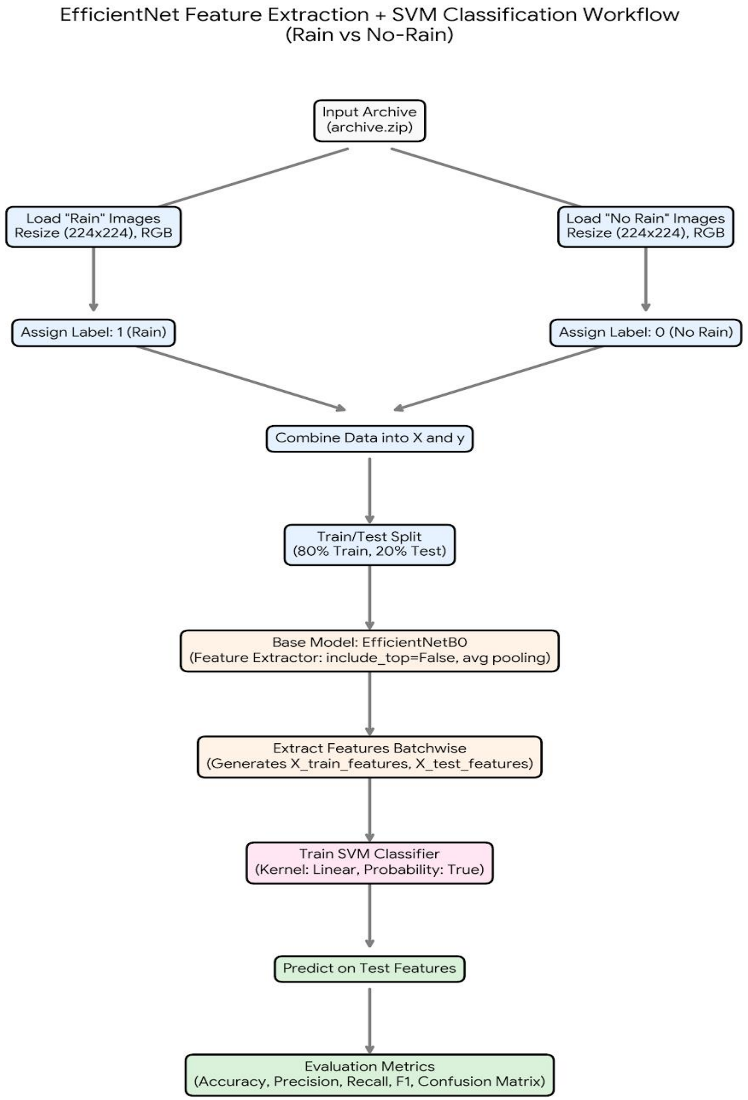
- **Advantage:**

1. Extracts deep visual patterns (edges, textures, clouds)
2. Faster than full deep learning and requires **less GPU computation**

VGG16 + SVM Pipeline



EfficientNet + SVM Workflow



Why EfficientNet + SVM

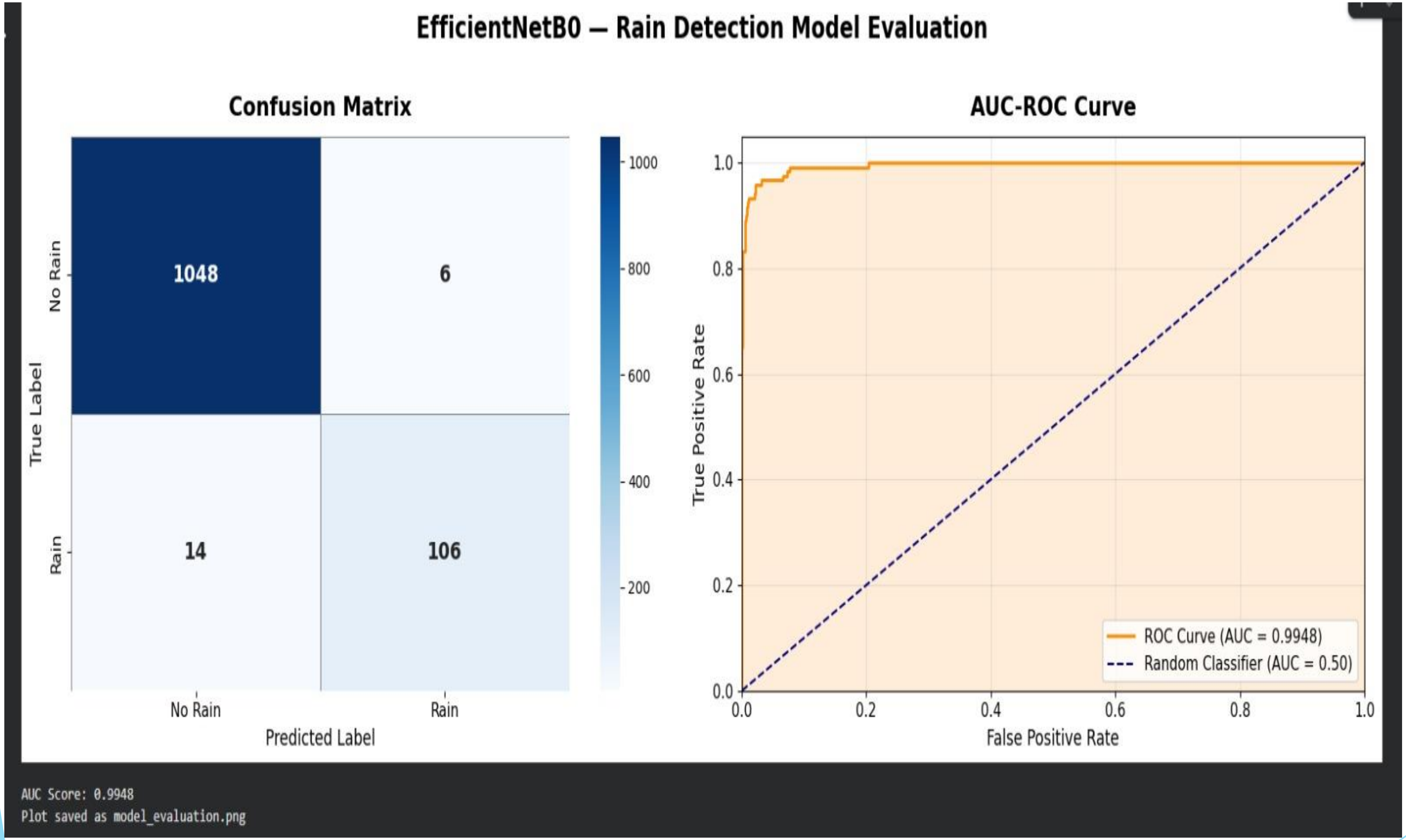
- EfficientNet extracts powerful deep image features automatically.
- SVM provides accurate classification using extracted feature vectors.
- Combination reduces overfitting and works well on medium-sized datasets.
- Achieves high accuracy and strong Rain vs No-Rain class separation.

Training Configuration

- Optimizer: Adam
- Batch Size: 32
- Epochs: 25
- Learning Rate: 0.001
- Dropout Regularization Applied

Experimental Results

Model	Accuracy / F1-Score
EfficientNetB0	98.30% / 0.9091
EfficientNet + SVM	98.10% / 0.9806
LSTM-Based Model	27.14%


EfficientNetB0 — Rain Detection Model Evaluation
Confusion Matrix
True Label
No Rain
Rain
1048
6
14
106
No Rain
Rain
Predicted Label
1000
800
600
400
200
AUC-ROC Curve
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC Curve (AUC = 0.9948)
Random Classifier (AUC = 0.50)
AUC Score: 0.9948
Plot saved as model_evaluation.png

Observations & Improvements

Limitations

- Dataset imbalance affects generalization
- Binary rainfall classification only
- Raw images are **high-dimensional and noisy**

Future Improvements

- CNN-LSTM hybrid models
- Attention-based cloud learning
- Multimodal weather fusion
- Real-time satellite forecasting

Approach 3: Time-Series Weather-Based Rainfall Detection

Soumadip Singha Mahapatra (2022ITB097)

Dataset - Kaggle

- **Historical Hourly Weather Data**
- Location: **Vancouver**
- Time Range: **2012-2017**
- Total Samples: **45,000**
- **Key Idea:** Convert raw weather data → time-series sequences for learning

- **Features:**

1. Temperature (Kelvin)
2. Humidity (%)
3. Pressure (hPa)
4. Wind Speed (m/s)
5. Wind Direction (degrees)
6. Weather Description (text label)

Data Preprocessing

- Missing values handled using **dropna()**
- **Label creation:** Rain keywords → Rain (1) and Others → No Rain (0)

- **Feature Engineering:**

1. Time encoding (sin / cos)
2. Lag features (t-3, t-6, t-12, t-24)
3. Rolling averages
4. **Min-Max** Normalisation

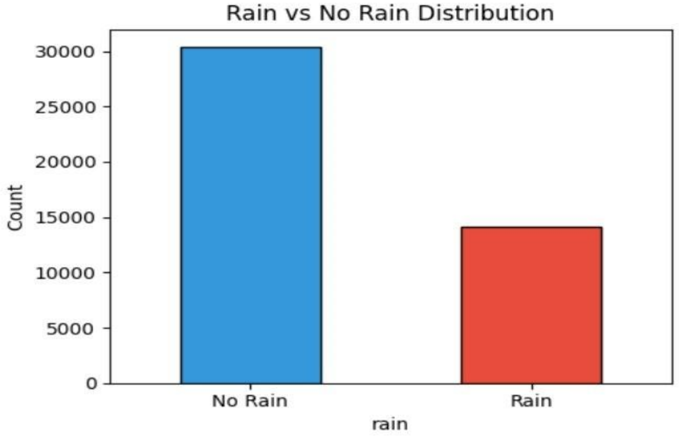
- Class balancing handled using **upsampling**

✅ Dataset loaded for city: Vancouver
Shape: (45253, 6)

[3]:

datetime	humidity	pressure	temperature	wind_speed	wind_dir	description
2012-10-01 12:00:00	NaN	NaN	NaN	NaN	NaN	NaN
2012-10-01 13:00:00	76.0	NaN	284.630000	0.0	0.0	mist
2012-10-01 14:00:00	76.0	NaN	284.629041	0.0	6.0	broken clouds
2012-10-01 15:00:00	76.0	NaN	284.626998	0.0	20.0	broken clouds
2012-10-01 16:00:00	77.0	NaN	284.624955	0.0	34.0	broken clouds

```
=== Class Distribution ===
No Rain (0): 30,389 (67.2%)
Rain    (1): 14,071 (31.1%)
```

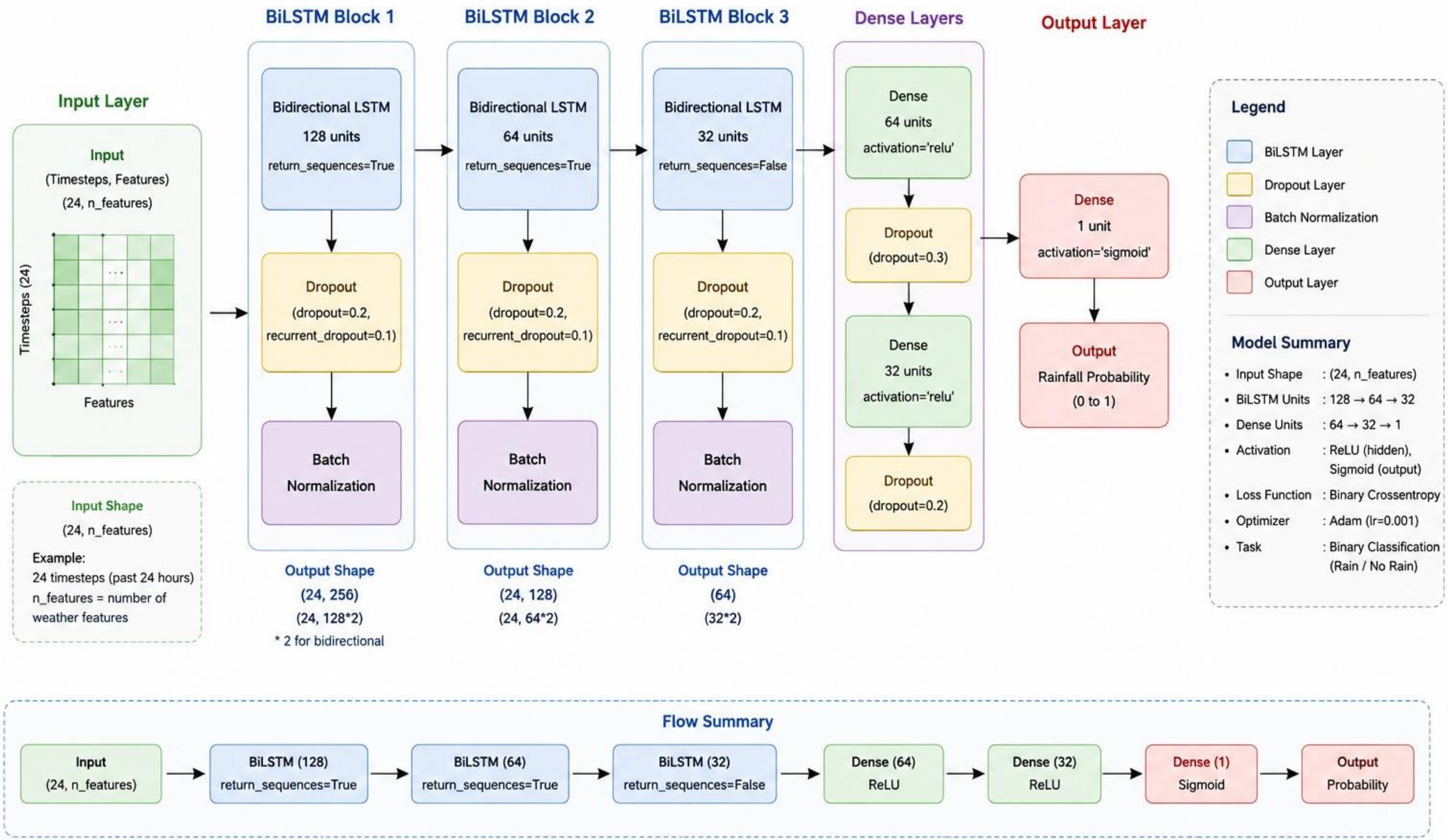


✅ Train : (39151, 24, 22) Labels: (39151,)
Val   : (8389, 24, 22) Labels: (8389,)
Test : (8390, 24, 22) Labels: (8390,)

```
=== Missing Values ===
humidity        1826
pressure        4234
temperature      795
wind_speed       795
wind_dir         795
description      793
dtype: int64
```

	humidity	pressure	temperature	wind_speed	wind_dir
count	43427.000000	41019.000000	44458.000000	44458.000000	44458.000000
mean	81.895480	1018.130915	283.862654	2.432746	159.892978
std	14.522221	15.792425	6.640131	2.004636	98.163221
min	12.000000	807.000000	245.150000	0.000000	0.000000
25%	73.000000	1012.000000	279.160000	1.000000	80.000000
50%	86.000000	1017.000000	283.450000	2.000000	140.000000
75%	93.000000	1022.000000	288.600785	4.000000	236.000000
max	100.000000	1100.000000	307.000000	25.000000	360.000000

BiLSTM Model Architecture



Note: Dropout and Batch Normalization are applied after each BiLSTM layer to improve generalization and training stability.

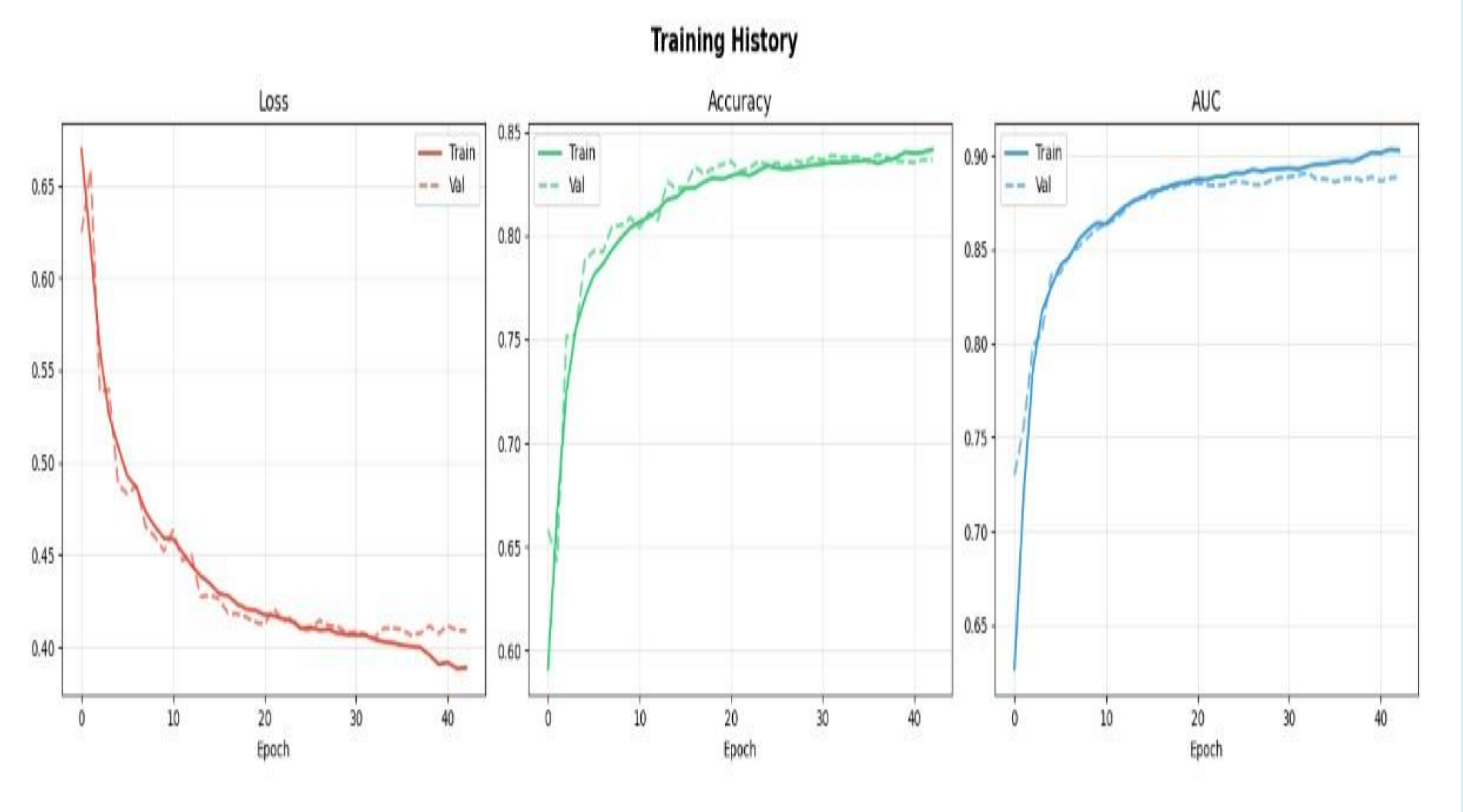
Training Strategy

Configuration

- Optimizer: **Adam** (lr=0.0001)
- Loss: **Binary Crossentropy**
- Batch Size: **64**
- Max Epochs: **60**

Techniques

- Split: **70/15/15** (Time-ordered, no shuffle)
- Early Stopping (patience=10)
- **ReduceLR** on Plateau (factor=0.5)
- Model Checkpoint (Save best **val_auc** weights)


Training History
Loss
Accuracy
AUC
Train
Val
Epoch
0.65
0.60
0.55
0.50
0.45
0.40
0.85
0.80
0.75
0.70
0.60
0.90
0
10
20
30
40

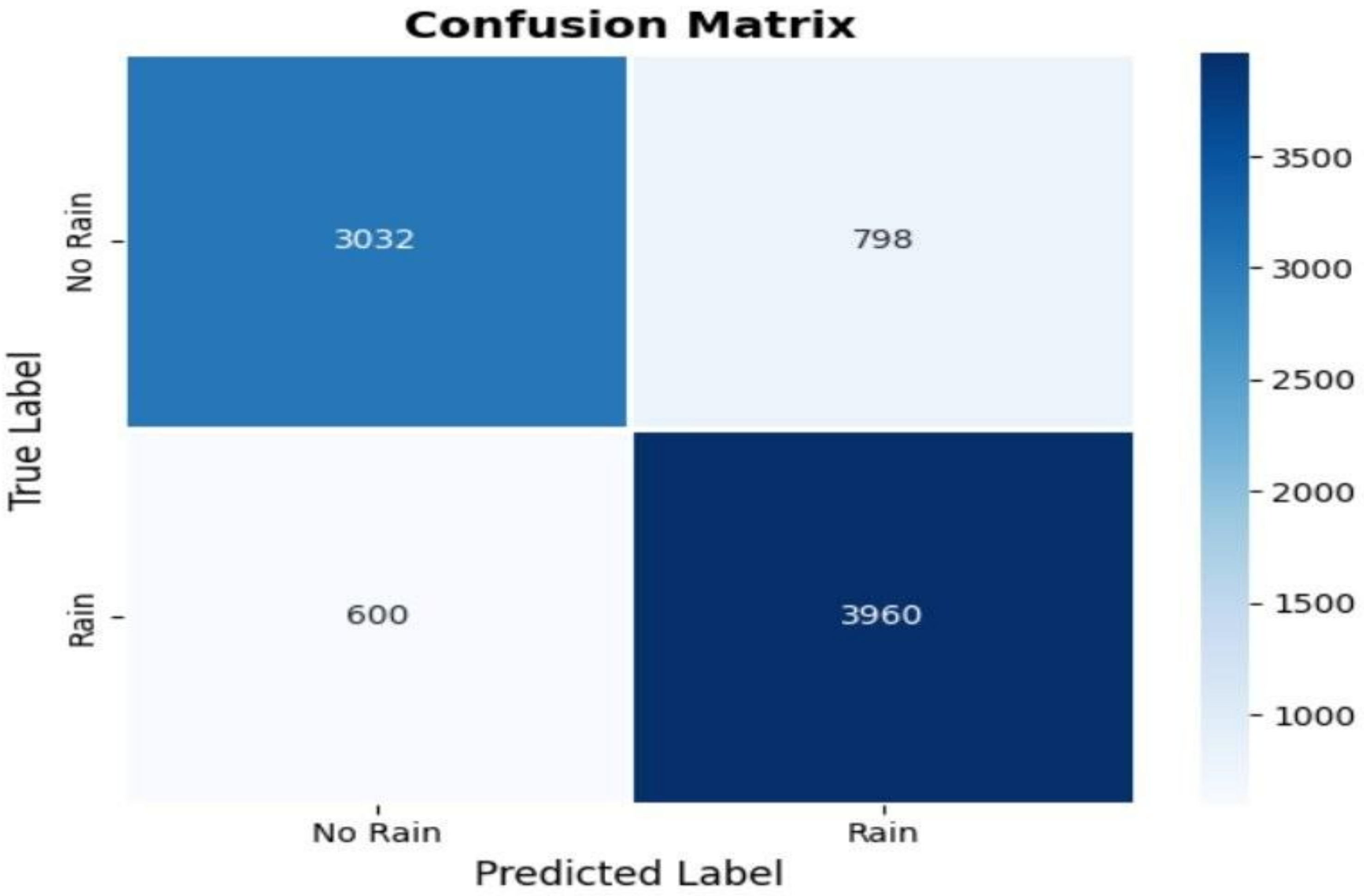
Results & Evaluation

Performance

- Accuracy: **83.34%**
- AUC: **0.8830**
- F1-Score: **0.8517**
- **Model correctly detects 87 % of actual rain events, critical for early warning systems**

Observations

- Balanced precision-recall
- Strong generalization
- Effective for time-series classification
- **BiLSTM captures:** Forward temporal patterns & Backward dependencies

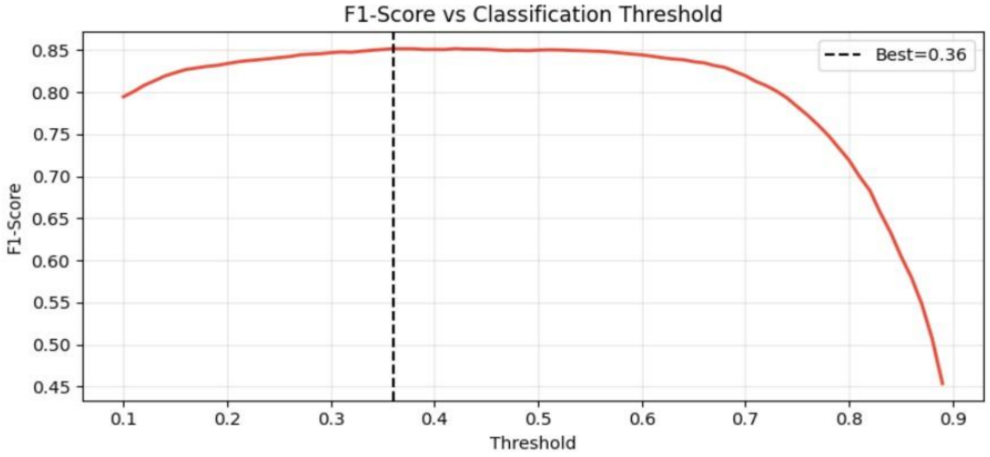


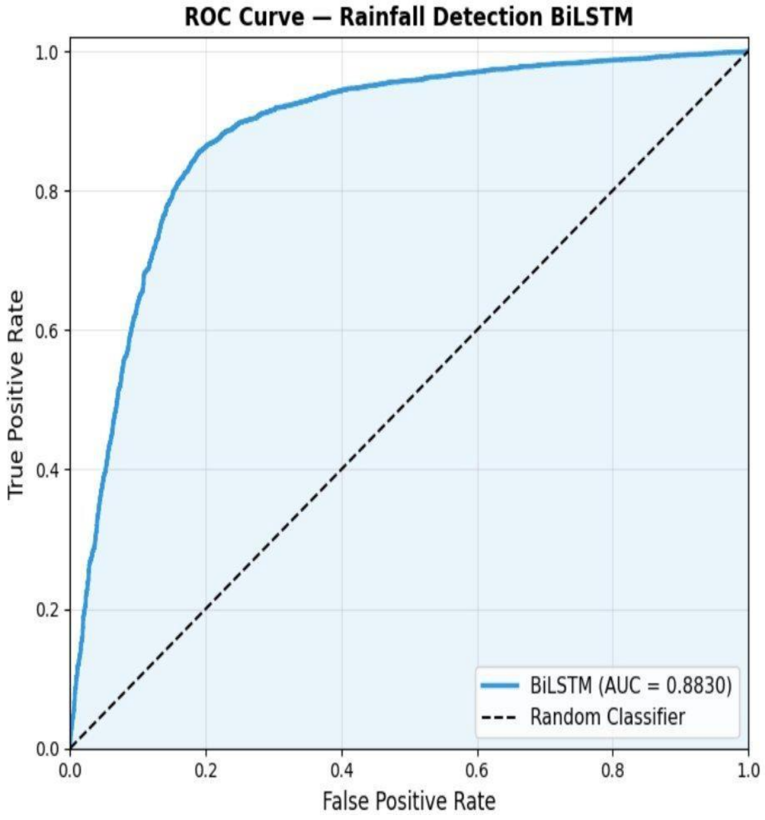

```
True Negatives  (No Rain predicted correctly): 3032
False Positives (Rain missed – said No Rain) : 798
False Negatives (No Rain – said Rain)        : 600
True Positives  (Rain predicted correctly)   : 3960
```

✅ Optimal threshold : 0.36
Best F1-Score : 0.8517

🔍 Classification Report with Optimal Threshold:

	precision	recall	f1-score	support
No Rain	0.86	0.75	0.80	3830
Rain	0.81	0.90	0.85	4560
accuracy			0.83	8390
macro avg	0.84	0.82	0.83	8390
weighted avg	0.83	0.83	0.83	8390





Overall Summary

- Rainfall prediction is a **complex, multi-factor problem** requiring diverse approaches, no single model can fully capture all atmospheric variations
- **Feature representation and data quality** are crucial for model performance
- Model performance is highly influenced by **data imbalance and preprocessing**
- **Combining complementary approaches** leads to more robust and reliable predictions
- Future systems should focus on **integrated, real-time, and scalable solutions**

Thanks

Any Questions?